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ORIGINAL ARTICLE

The development of an entrepreneurial mindset in primary education

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Abstract

This article provides insights into the design thinking method, a teaching strategy for developing creativity, innovative thinking, and an entrepreneurial mindset in young people as a universally applicable skill. It presents results from a study of 146 seventh- and eighth-grade students, aged 12–14 and 20 teachers across 10 Slovene schools. Teachers used the design thinking method to facilitate hands-on student projects and evaluated their effectiveness in cultivating an entrepreneurial mindset. The two teachers per class submitted a combined final report of their experience and observations. In addition, a study facilitator visited each school several times to discuss the project with the teachers, observe the class, and take notes. Thirteen factors that contributed to students' entrepreneurial mindset were identified and organized into three clusters: project factors, the learning environment in which it was carried out, and the learning and teaching factors.

1 | INTRODUCTION

This article focuses on the development of young people's "entrepreneurial mindset" which McGrath and MacMillan (2000, p. 32) defined as "the ability to sense, act, and mobilize under uncertain conditions." In 2012, the European Commission released the report *Entrepreneurship Education at School in Europe: National Strategies, Curricula, and Learning Outcomes* which outlines strategies to enhance creativity, innovation, and entrepreneurship at all levels of education and training in 31 European countries (Bourgeois, 2012). It highlights the *Republic of Slovenia's 2005 Development Strategy* (SDS) which states that "the entrepreneurial spirit of young people should be encouraged … [and] training programmes for running small businesses and entrepreneurial knowledge for young

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people should be provided" (Bourgeois, 2012, p. 71). In Slovenia, entrepreneurship is optional for upper secondary school students (university-bound students aged 15–18). Slovenia has established learning outcomes in primary education (age 6–14) that are associated with economic and financial literacy. Additionally, developing an entrepreneurial mindset is covered in the compulsory subjects of maths, sciences, and technology. Although teachers' awareness of entrepreneurial education has grown and become more positive, many lack the training and continuing professional development to teach this mindset (Seikkula-Leino, Ruskovaara, Ikavalko, Mattila, & Rytkola, 2010). It is often regarded as only necessary for an entrepreneurial career, even though studies have shown that school activities that develop practical knowledge and skills are applicable anywhere in society, even if people seek wage employment rather than self-employment (Kucel, Róbert, Buil, & Masferrer, 2016).

This study argues that design thinking can be an effective method of developing the entrepreneurial mindset. It provides a model for the training that students are missing and observes the outcome of teachers who put that training into practice following an elective course offered by 10 Slovenian primary schools.

2 | DEVELOPING THE ENTREPRENEURIAL MINDSET

An entrepreneurial mindset has been identified as important to increase competitiveness (Brorstrom, 2002) from the firm's, the municipality's, or the country's perspective (Jury, 1999). Several scholars have provided broad definitions, but the most widely used are McGrath and MacMillan's (2000, p. 15) and Ireland, Hitt, and Sirmon's (2003, p. 968) who defined it as a "growth-oriented perspective through which individuals promote flexibility, creativity, continuous innovation, and renewal." Ireland et al. (2003) identified several components. The first is the ability to recognize opportunities, which, in its broadest sense, encompasses identifying information asymmetries and developing different beliefs about the relative value of resources once transformed into products or services (Shane & Venkataraman, 2000). The second is entrepreneurial alertness, an ability to identify valuable goods and services which, at a certain time, provide unexpected value. The third is the use of real options logic, an approach to resource allocation which follows three general rules: sequencing, low commitment, and reallocation (Klingebiel & Adner, 2014). The fourth and final component is the use of an entrepreneurial framework which includes establishing an opportunity register, goal setting, and determining the right timing to exploit an entrepreneurial opportunity.

The "entrepreneurial mindset" can be analyzed from two perspectives. In relation to commercial activities, it involves groups and individuals who create marketable products and services or are entering different phases of establishing and managing a company. These activities hardly overlap with the educational activities of young people in primary education. The second position, which has been gaining in prominence in the last decade turns the spotlight on the promotion of creative and innovative problem-solving as a universally applicable skill (Neck & Greene, 2011; Sarasvathy & Venkataraman, 2011). Some scholars argue that teaching these skills is one way of preparing students to face the unpredictable societal and labor market demands of the decades ahead (Murgatroyd, 2010; Sahlberg & Oldroyd, 2010). Regardless of the strong consensus that developing an entrepreneurial mindset is necessary for a successful career and national competitiveness, there is no consensus on how to develop it. Attempts include using serious games among engineering students (Bellotti et al., 2014), developing skills that are inherent to entrepreneurs (McGrath & MacMillan, 2000), and using best practices from teaching entrepreneurship in the creative arts (Pollard & Wilson, 2013).

Reports on using specific methods to develop an entrepreneurial mindset at the pre-university level are very scarce, yet some scholars have argued that childhood and adolescence were the ideal times to teach entrepreneurial skills and foster a positive attitude toward entrepreneurship (Seikkula-Leino et al., 2010). Löbler (2006, p. 26) argues,

> If we look at the "skills" and competencies of entrepreneurs from a constructivist's perspective we find most of them by observing children under the age of five or six: they are motivated to learn, they are interested in a variety of different topics, they ask excellent questions, they try many things to get insights, they are creative, they are impatient.

Additionally, in his book, *In the Beginning was Entrepreneurship*, Johannisson (2010) suggests that children are born enterprising, but this is not encouraged while growing up.

3 | THE DESIGN THINKING METHOD

Curricula that are designed to teach creative and innovative problem-solving skills provide students with a generally applicable skill for their personal and professional life (Meinel & Leifer, 2011). One attempt that is underway to operationalize such curricula is the design thinking method (Figure 1) which promotes the development of creativity, innovation, and an entrepreneurial mindset with a strong focus on understanding the needs and desires of the end user (Meinel & Leifer, 2011; Rauth, Köppen, Jobst, & Meinel, 2010). It requires various data collection techniques. After collecting sufficient information about users and their wishes, desires, and challenges, the problem to be solved is then defined and ideas are generated according to different criteria. Next, the prototyping process involves different methods to visualize, create, and test the solution to the problem. This process is intended to be cyclical and iterative, as some steps in the process or the entire process are repeated to improve the product.

Students who are taught design thinking find themselves in situations where they must consider the social context, balance different and even contradicting information, and act responsibly. These situations challenge them in ways that go beyond the traditional scope of school curricula, as they involve the wider community, cooperation with external stakeholders, and the testing of solutions with real users. This article presents the results of a study that evaluated the effectiveness of the design thinking method in elementary schools in Slovenia.

4 | METHOD

To identify the potential of design thinking as a method to teach the entrepreneurial mindset to young students in Slovenia, we monitored the progress of a project in selected elementary school classrooms, each with 10–20 students and 2 teachers. We also investigated the implementation of the projects in the participating schools, how they were connected to their local communities, and whether they enabled the students to learn in real-life situations. This study used ethnography to examine the experience of teachers in facilitating a design thinking project in their classroom. The research questions were:

1. How did teachers use design thinking as a methodology for teaching the entrepreneurial mindset to adolescent students?
2. What was the context in which the teaching took place and which factors influenced the completion and outcomes of the projects?

The project took up two hours of classroom time per week for 17.5 weeks (35 hr). A total of 146 seventh- and eighth-grade students aged 12–14 and 20 teachers (2 per class) from 10 Slovene schools (labeled A–J in the text)

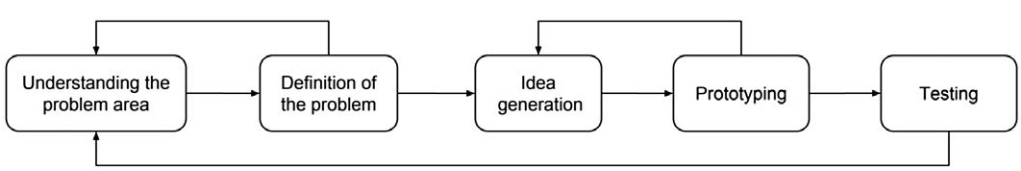

FIGURE 1 The design thinking method.
Source. Carroll et al., 2010; Rauth et al., 2010

took part in the study. Teachers self-selected their classes and the assignment was part of an extra-curricular activity. Parental approval for students' participation was obtained. The purpose was to use the design thinking method to develop the entrepreneurial mindset in students by training them in creative and innovative problem-solving that focused on the end user. Teachers were given instructions and asked to lead students through the process which was divided into five interrelated phases. The first was *understanding and defining the problem*. Students identified and defined a local or social problem that could be solved with a new product, service, or other solution. The second was *observation* where students observed and spoke with people who were affected by the problem and gathered information using different methods, such as interviews and videos. The third was *ideation*. On the basis of the data collected, students brainstormed ideas that would help to solve the problem. The fourth was *prototyping and testing* in which students created a solution using different prototyping methods such as metal or woodworking, creating a report or exhibit, and organizing an event. The students then tested the prototype with real users or role-playing users when real users were unavailable in order to determine if the solution brought the expected benefits. The final phase was *implementation* where students brought the solution to its completed stages. If time and resources allowed, they could put the solution to use.

To train in the design thinking method and ethnographic research in preparation for the classroom project, the participating teachers attended a two-day seminar facilitated by the researchers. They were instructed in creativity, the entrepreneurial mindset, current and best practices and the design thinking method, as well as in field observation and note-taking. They were taught to take regular notes on progress during the course of the project, focusing on its phases. They also had to describe their role in managing the project and highlight the problems they encountered. Using this ethnographic approach, we gained a deeper insight into the circumstances of the project and teachers' interaction with students and external collaborators.

During the project, the teachers attended two meetings to present their progress. In addition, one of the study facilitators visited each school several times to discuss the project with the teachers, observe the class, and take notes. The two teachers per class submitted a combined final report of their experience and observations. A total of 10 teacher reports and 10 researcher notes from class visits were analyzed. The information and presentations were compared with the submitted written reports and no discrepancies were found, thus supporting their credibility. Using a grounded theory approach (Strauss & Corbin, 1997), three raters coded the reports. The codes were first categorized according to the design thinking phases of the project. Additionally, 13 themes that were based on repeated or interesting ideas, concepts, and keywords were identified throughout the coding process. They were divided into three sets of factors that contributed to the development of students' entrepreneurial mindset: project factors; environmental factors; and learning and teaching factors.

5 | RESULTS

The results presented here are organized according to the design thinking phases and supported with excerpts from the teachers' reports. We discuss the three factors that contributed to the development of students' entrepreneurial mindset.

5.1 | The design thinking phases

5.1.1 | Phase 1: Understanding the problem area and Phase 2: Defining the problem

Each of the participating classrooms developed an innovative idea to address a problem in their school, social circle, or community. First, in phase 1, schools had to find an unmet need and in phase 2 define a challenging but focused problem to tackle, thus meeting the identified need. The classrooms therefore undertook field observations, some

in their school and some in the local community. Some students were also offered the opportunity to return to their field location more than once or make observations in their own free time. School H reported, "The students had one week to observe their school and its surroundings." The understanding phase supported the intended learning outcome of understanding different ways to observe, compare, and analyze people's needs. School D reported that the students were able to learn about the problems of the locals by interviewing them and assess the feasibility of solving the observed problems with the teachers' help.

The observation phase supported the intended learning outcome of personal qualities that were necessary for the successful realization of ideas, such as creativity, self-confidence, enthusiasm, courage, and cooperation. Students seemed very enthusiastic, active, and creative during this phase. School G reported, "We came up with 240 different identified needs to draw projects from." Additionally, as school J reported, they did not have to be coerced because they volunteered for the observation and data collection work. Following the field observations, they began the ideation phase in which they brainstormed ideas to address the identified problem.

5.1.2 | Phase 3: Ideation

The ideation phase was one of the most engaging phases of the project. School C reported, "The students had so many suggestions, but we had to postpone them until further notice." During this phase, the classrooms were successful in identifying one feasible project each on which to work. For example, in School B, which addressed the problem of an underused forest trail, the students organized a science day and installed three additional educational displays on the trail. School J, which identified the need for a "school gift," chose to create a soap using local ingredients. School I developed an audiobook because a relative of one of the students suffered from dyslexia and the students sought ways to help that relative. In some cases, the teachers were able to obtain outside assistance, which enhanced the project. In Schools E and H which addressed the problem of their school's poor-quality lockers, the students worked with a local carpenter to design and build a prototype of a new locker. Table 1 provides a description of each school's problem and chosen solution.

During this phase, teachers facilitated the students' negotiation process of choosing a few ideas from many suggestions. School C, which worked on a clothing line, reported, "We agreed that every group had to incorporate as many ideas as possible into the clothes, even the most unusual and 'crazy' ones." The experience supported the learning outcomes of developing interpersonal communication and learning to take risks, negotiate, and compromise.

5.1.3 | Phase 4: Prototyping and testing

Once each class had identified a problem and a potential solution, the students worked on creating a prototype of their proposed solution. During this phase, some classes conducted additional research and further explored the technology for creating the product. For example, School J, which chose to create a soap, reported that "the students tried out quite a few possibilities. They worked in teams and felt no apprehension of possible failure." In some cases, students approached external collaborators for resources they did not have. School B reported, "We asked external collaborators to provide us with equipment." Students collaborated with teachers from other subject areas or external partners such as entrepreneurs, designers, and university professors to create their prototype. School C, which worked on a clothing line, collaborated with a local fashion designer who came to the school and provided mannequins. It reported, "Our ideas were slowly becoming applicable, we managed to create prototypes from affordable materials." The students decided to make women's dresses and handbags using a "collage" technique that incorporated metal and materials from seatbelts.

Teachers reported prototyping approaches ranging from the use of simple materials that were manipulated by hand (e.g., cloth and wood) to more complex technologies such as 3D modeling. School F, which worked on designing sports equipment, reported, "The students first sketched their ideas on paper and then produced 3D

TABLE 1 List of all projects

School	Problem	Project
A	The school has no identity	School mascot
B	Underutilized forest trail	Science day and installation of three additional educational displays
C	People want to stand out with their outfits	Clothes and fashion items made of recycled car accessories, a fashion show
D	There are no local souvenirs or tourist items	A postcard
E	Poor quality school lockers	Design and build a prototype of a new locker
F	An empty classroom	Sports equipment and design of a classroom where students can exercise during breaks
G	Lack of tourist services at a farm	Development of additional services (e.g., renting bicycles, swimming pool, Segway tours, and a forest trail)
H	Poor-quality school lockers	Design and build a prototype of a new locker
I	Students with dyslexia have a difficulty reading books	Audiobooks
J	Need for a "school gift"	Soap using local ingredients

models of the cupboards using Google Sketch Up." The prototyping phase was one of discovery. School J reported, "The students were surprised by the change in smell and colour of the soap when adding fragrances and spices." It was also a project highlight for some of the classrooms. School H reported that, compared to the other phases, the students were most enthusiastic about their interaction with practical, hands-on work.

Once students had completed the prototypes, they tested them with their target users, including their peers, guests they visited during the field observations, and tourists. School D reported, "We designed a prototype of the [postcard] and asked parents, teachers, our headmaster and visitors to evaluate it." Some schools organized multimedia presentations of their products for their users. School E, which developed prototypes of new lockers, reported, "Each group presented their prototype with a short video and tested its applicability with users." Some students also changed their prototype during the testing phase to meet users' wishes, thus opening up new possibilities and ideas. School G, which worked on marketing materials for a tourist farm, reported, "Test results gave rise to new ideas for upgrading and improving the prototypes." Some schools also sought feedback from external collaborators on how to present their products to users. School J reported, "The head of the company Dima Center explained the significance of marketing and added value of a product."

5.1.4 | Phase 5: Implementation

During the implementation phase, the schools finalized their products, presenting them for assessment or in some cases putting them to use. School H, which developed new school lockers, reported that the students presented their idea from conception to realization to the school leadership. Some schools even presented their results to community leaders. School E reported, "The students published their results in various media and presented them to municipal representatives." Some schools also sold their products at a school bazaar (School D) or other venues. School C, which developed a clothing line, reported, "We organised a fashion show in the centre of Celje and offered our clothes to be sold at a charity auction." Some schools found that this was one of the best ways to understand users and their needs. School I, which created an audiobook for challenged readers, reported, "It was nice to assume the role of sellers and see how they [customers] think."

One unanticipated finding was the many examples of schools' ongoing relationships with external collaborators after the project. School B, which worked on upgrading a forest trail, reported, "We will organise guided tours along the educational path for seventh-grade students of neighbouring schools. We will also join forces with the local Tourist Association and the Pensioners' Association." School F, which developed sports equipment, presented its prototype to a local company which was interested in further cooperation and perhaps even its commercial manufacturing.

5.2 | Factors that contributed to the development of students' entrepreneurial mindset

Based on the analysis of the reports which highlighted situational and contextual learning, we identified three sets of factors that contributed to the development of students' entrepreneurial mindset, as listed in Table 2: project factors (4 items); environmental factors (5 items); and learning and teaching factors (4 items).

5.2.1 | Project factors

The analysis indicated that the projects of all 10 participating schools involved field work (1st project factor). The problems on which the respective schools worked were closely connected with the lives of their local communities. The students often sought solutions to their own problems or those of their peers. Some schools transferred the education process to their respective partners' offices or manufacturing halls; some partner firms also invited their partners to the education process.

The role of the teachers in facilitating the project was critical (2nd project factor). First, some schools noted that requiring teams of two teachers was helpful. School H reported, "Two mentors working together was an excellent idea." Second, schools indicated that the teacher training for the project was critical and that they needed the support of their peers, mostly because of the unpredictability of the projects. Teachers from School C reported, "Although we carefully planned the work for the project, the events sometimes took a different turn." Third, teachers' enthusiasm was important as it affected the whole group. A teacher from School J reported, "We encouraged participants by setting examples, crazy ideas, playfulness, humour and the feeling of freedom in the group."

The third factor was allowing students to choose relevant and meaningful projects. A teacher from School H reported, "My role as teacher was different [from how I normally teach], because we wanted the students to find the best projects by themselves." Reports suggested that the hands-off approach was important for the success of the design thinking method in classrooms because it allowed students to develop skills and competencies such as inquisitiveness, creativity, innovativeness, self-confidence, and perseverance. If these projects had been overly

TABLE 2 Factors that contributed to the development of students' entrepreneurial mindset

Project factors	Environmental factors	Learning and teaching factors
• Field work and the duration of the project • Trained and enthusiastic teachers • Relevant and meaningful problems to work on • Interactive, hands-on projects	• External approval from school, leadership, parents, and community members • Opportunities to work with external collaborators in an authentic environment • Adequate infrastructure, including time, resources, and external assistance • Applied use of new tools, resources, and technologies in the classroom, particularly prototyping tools • Testing prototypes in the environment	• Student-centered education and including all students in the work • Team work and an interdisciplinary approach • Studying different options and assessing ideas critically • A culture of trust and safety

directed by adults, particularly in the processes of selecting the problem, solution, and prototyping approach, students would not have felt ownership of the products and may have become disenfranchised. With deliberately limited teacher involvement, students may experience failure in developing a practical solution, but this is common in real life and a valuable lesson for students. Therefore, teacher training is important to help teachers understand their role in facilitating the projects and students to negotiate ideas in groups and address both successes and failures.

The fourth factor was interactive, hands-on projects where students were encouraged to develop interactive projects. Hence, teachers primarily supported those where students could develop the solution by themselves.

5.2.2 | Environmental factors

The second set of factors, collectively named *environmental factors*, is the effects of the environment, especially as they relate to the support of the school leadership, parents, and the local community. The first factor in this group is external approval. All schools reported that they presented their project to their school's management, parents, and local authorities. Several publicized their projects in local newspapers, on school websites, and even on local radio and TV stations. School C organized a public fashion show and school J used the product to offer it as a gift to several other institutions on various occasions.

According to reports and observations, all schools cooperated with a number of external institutions and collaborators (the second factor within environmental factors). The students and their teachers visited local companies and entrepreneurs and worked with other local schools and experts. For example, the students of school A presented their prototype to the local community and arranged further collaboration. Collaboration with outside partners and the local community provided the students with many opportunities to discover new challenges and create fresh ideas. This also developed social capital, which is an important component of lifelong learning (Kilpatrick, Field, & Falk, 2003) and learning performance (Žakelj & Ivanuš Grmek, 2013).

An adequate infrastructure and the applied use of tools and technologies are the third and fourth factors of environmental factors. These were necessary to carry out all projects, especially in the prototyping phase. The reports suggested that the schools possessed adequate spatial and material means for the project, thus upholding the notion that their physical learning spaces were aligned with such pedagogical approaches (van Merriënboer, McKenney, Cullinan, & Heuer, 2017). In many cases, the students brought some of the necessary equipment and materials from home and external collaborators provided the rest. In some schools, the educational process took place on the premises of external collaborators. Students used a variety of prototyping techniques and often developed the prototypes in their spare time at home or at a local workshop. When the prototypes were created, they were tested with different users (fifth environmental factor). Prototypes were also an important means of communication with the local environment, including parents, school management, and local media. Testing was carried out with users, sometimes in school grounds and often in the field, as with School B where testing took place in the local forest.

5.2.3 | Learning and teaching factors

The third decisive set of factors that meaningfully contributes to an educational environment that is suitable for developing an entrepreneurial mindset in primary education is the teaching and learning methodology. The first two factors are an inclusive culture and teamwork where students form a cohesive team that welcomes and takes advantage of each member's strengths. One student is good with prototyping tools, another with multimedia presentations and a third with organizing team work. It is important that each student feels part of a team working toward a common goal. With such multi-faceted and multidisciplinary problems, schools reported that this was easier to achieve than with traditional classes where all students worked on the same tasks. There were several activities where all students needed to participate, such as brainstorming, but others were also distributed within the group. The third

factor is being open to different options and ideas. School H reported that the students were very enthusiastic about the practical work where they combined ideas from several sources. "We complemented our ideas with those of a fashion designer, which felt really great," reported school C. Students also learned by exploring different options and making mistakes in the prototyping phase. School J, which decided to create a soap, reported that "the students were surprised to witness the change in fragrance and colour as they added incenses and spices. They worked in teams and were not afraid of experimenting and making mistakes." This attests to the importance of the culture of trust and safety as the fourth factor. Creating this culture allows for risk and even encourages making mistakes and learning from them, which is not typical of most normal classes. At the same time, however, the creative culture offers plenty of opportunities for the engagement of the individual, collaboration, and the construction of knowledge.

6 | DISCUSSION

From the point of view of the development of creativity and innovation in school, the knowledge taught in all subjects is important. However, the world is not split up into school subjects—it is an undividable whole, which is why collaboration between different subjects is imperative. The complexity of the problems which students faced required an interdisciplinary approach and collaborative learning. The interdisciplinary nature of the work was manifested in several ways and was both observed and evident from teachers' notes. Classes were conducted by two teachers and, in 8 out of 10 cases, these teachers came from different disciplines. For instance, at school C their background was in arts and home economics. Additionally, students and teachers in all cases invited guests to help them with the knowledge and skills needed for the successful completion of the project. In school B, they invited a biology and a history teacher; in school C, students worked with a parent who was a fashion designer and in school G, they collaborated with several people working at the tourist farm.

Teachers reported that the participating students were focused on the process rather than on the product itself so their work and learning were to a large extent self-regulated, meaning that they were active in a metacognitive, motivational, and behavioral way. Studies show that self-regulated students, in contrast to peers who are not self-regulated learners, are highly motivated (Elstad & Turmo, 2010; Kolovelonis, Goudas, & Dermitzaki, 2011). They are more confident and more involved during classes and use different learning strategies. In addition, the opportunity for creative work, the readiness for risk-taking, and putting great effort into work grow through self-regulation, which an entrepreneurial mindset helps to foster.

The next feature of the development of an entrepreneurial mindset in primary education is the contextualization of learning which stresses that the learning process happens in interaction and collaboration with the social, societal, and cultural contexts (De Corte, 2000). Our results show that the participating students had many opportunities to discover new challenges and come up with fresh ideas. Collaborating with the local community enabled them to better understand what went on in the community. The programs and activities were developed in dialogue with the community and with various stakeholders. The experiences gained made the transition from schooling to adult life much easier for adolescents, helped them become acquainted with the world of work, influenced the development of independence and responsibility, and provided a sound basis for further education and work. Besides the development of social capital, which is an important component of lifelong learning, there is also great potential for innovation at this level (Kilpatrick et al., 2003). Many companies and organizations are aware of this, as they understand that these young students represent their potential hiring pool.

The characteristics of the curricula, based on the use of design thinking to help students engage and solve authentic societal and business problems, adhere to the basic features of an effective learning process according to De Corte (2000). It is constructive, as knowledge is acquired through contextualized experiences that are self-regulated, cumulative, goal-oriented, situated, and collaborative and differs from individual to individual. The design thinking method balances extensive options for discovery learning and personal exploration with systematic guidance by teachers who are masters of the problem-solving process. Students become agents of their own

learning, as learning takes place in a personally meaningful, authentic context that supports individual differences among learners.

Developing an entrepreneurial mindset should not, however, be understood as being at odds with the established teaching routines and the existing curricula. The challenge is that teachers, as thinking practitioners, need to build bridges between the demands of the official curriculum and the learning environment that are necessary to develop the entrepreneurial mindset. Some scholars believe that it is only in the school that serious progress can take place (Darling-Hammond, Ramos-Beban, Padnos Altamirano, & Hyler, 2015; Istance & Dumont, 2013). This may sometimes require a radical change in the learning environment and the associated approaches to teaching and learning, but not in the curricula itself. To answer the question of how to design powerful and motivating learning environments, adequate proof and best practice examples are necessary. On their own, best practice examples such as those observed in this study do not yet constitute sufficient ground for redesigning schools and school policies, but they do send a powerful message about what drives and what hinders learning.

While the design thinking method as it was implemented in this study yielded very positive results, it is not without challenges. Teachers reported and we observed several pitfalls that should be taken into account to minimize potential negative effects on learning. First, there must be a climate of confidence and security where risks and errors are normal constituents of learning and teaching. For instance, when investigating user needs, students will be exposed to personal stories and data or business-related confidential information that is not to be shared, sometimes not even among classmates. For instance, students working with the tourist farm came into contact with confidential business-related information and were asked not to share it with the general public.

Additionally, facilitating these projects required time and resources beyond what is perceived as normal within a school environment. Field trips, including transport and food, are sometimes a heavy burden for schools and parents. Teachers were required to work extra hours to facilitate these projects and extra resources within school budgets had to be allocated to support these activities, especially if two teachers were present at the same time. In this case, schools could apply for additional government funding for novel teaching approaches so that they could fund these activities. However, these approaches to funding the programs are not sustainable because they are not guaranteed from year to year.

Lastly, whilst failure is an important part of the experience, it is nevertheless difficult for teachers, school leaders, and students alike. Some projects will not be sustainable after the class ends or, in some cases, students will realize at mid-project that their product is not technically feasible, financially viable, or desirable within the target group. This happened in two cases among the 10 schools that took part in this study. School G learned shortly after it presented its solutions to the tourist farm that none of the suggestions would be implemented due to unexpected financial hardship. Similarly, one of the schools (School H) that was developing upgraded school lockers was faced with a lack of resources to build the improved lockers. An important lesson here is that the solution to a problem that students are working on should not be overly dependent on third parties and that teachers should be prepared to start a new project as time and resources allow.

7 | CONCLUSION

As is evident from the description of the classes that participated in this study, they were not strictly associated with the individual subjects that are traditionally taught in primary school, but focused on the larger thematic whole and on developing broadly applicable problem-solving skills. Such approaches are gaining in popularity. In Finland, for instance, traditional school subjects are being successfully replaced by topic courses which are interdisciplinary in nature (Halinen, 2015; Niemi, Toom, & Kallioniemi, 2016). This approach contributes to the integration of knowledge where subject areas are not excluded, but are linked and complementary. Additionally, such knowledge and skills are then more easily transformed into skills that are needed in everyday life as students develop and train them by solving authentic problems in everyday situations.

This study indicates that the design thinking method seems to be an effective pedagogical approach to develop the entrepreneurial mindset in elementary schools. It is, to the best of our knowledge, the first study of this scale and on this age group. The experiment that is documented in this article was not intended to revolutionize the schooling system and transform all subjects. Rather, it was an experiment in helping students develop the entrepreneurial mindset and competencies that many are not learning in their subject classes or in their daily life. Much work remains before traditional curricula are updated. However, several successful attempts at applying novel approaches to more traditional subjects have already been documented by researchers. For instance, design thinking has been applied to upgrade geography courses (Carroll et al., 2010).

Our findings indicate that an entrepreneurial project, such as the one presented here, could be used to meet certain currently recommended academic outcomes in the elementary school curriculum. Second, the development of creativity and innovation as the basic components of the entrepreneurial mindset in young people requires a systematic and comprehensive approach. Third, implementing high-quality education aimed at developing the entrepreneurial mindset calls for a comprehensive framework and the identification of effective practices and support from local and regional bodies. However, as this study suggests, elementary schools can relatively easily incorporate contextualized problem-based teaching practices aimed at developing an entrepreneurial mindset on their own without any major changes at the national or global levels where the chances of significant progress in the near future seem unlikely. Initial changes do not need to be made at the uppermost levels, but can be incorporated from the bottom-up through teacher education and initiative.

This study has many strengths but also a few limitations. First, we relied on disparate teacher reports of student behavior and learning. Different teachers may assess the success of this assignment differently, thus clouding the results. However, since individual teachers do all the assessing of their students in all areas of instruction, it is appropriate to rely on them as accurate assessors of their students' progress. Second, this assignment was tested on only one age group: 12–14-year-olds. Thus, these results are limited to generalizing within this age group. Third, teachers self-selected their classes and may have biased the sample by choosing students that they thought would enjoy or excel under the design thinking method. Future studies can assess the effectiveness of such a program at younger and older ages. Future research can also assess whether repeating this type of assignment at multiple ages can further entrench this way of thinking, or whether there is an ideal age at which to teach the entrepreneurial mindset to achieve maximum effects. Additional need for further research is to take students who have been through one of these programs and compare them with a control group that has not had the same opportunity to confirm that students who participate in the design thinking class are indeed developing entrepreneurial mindset above and beyond the control group.

To achieve the desired results in the classroom, it is very important to train teachers and school management teams so that they support and create learning environments with a "high degree of certainty" (De Corte, 2013) and thus alter the positions on and convictions about teaching and learning. As we have shown, the entrepreneurial mindset can effectively be taught to adolescents, engendering these valuable skills in people's formative years and preparing for greater entrepreneurialism in society as these students enter the working world.

REFERENCES

Bellotti, F., Berta, R., De Gloria, A., Lavagnino, E., Antonaci, A., Dagnino, F., & Mayer, I. S. (2014). Serious games and the development of an entrepreneurial mindset in higher education engineering students. *Entertainment Computing*, *5*, 357–366. https://doi.org/10.1016/j.entcom.2014.07.003

Bourgeois, A. (2012). *Entrepreneurship education at school in Europe. National strategies, curricula and learning outcomes*. Brussels: Education, Audiovisual and Culture Executive Agency.

Brorstrom, B. (2002). The world's richest municipality: The importance of institutions for municipal development. *Journal of Economic Issues*, *36*, 55–78. https://doi.org/10.1080/00213624.2002.11506443

Carroll, M., Goldman, S., Britos, L., Koh, J., Royalty, A., & Hornstein, M. (2010). Destination, imagination and the fires within: Design thinking in a middle school classroom. *International Journal of Art & Design Education*, *29*, 37–53. https://doi.org/10.1111/j.1476-8070.2010.01632.x

Darling-Hammond, L., Ramos-Beban, N., Padnos Altamirano, P., & Hyler, M. E. (2015). *Be the change: Reinventing school for student success*. New York: Teachers College, Columbia University.

De Corte, E. (2000). Marrying theory building and the improvement of school practice: A permanent challenge for instructional psychology. *Learning and Instruction*, *10*, 249–266. https://doi.org/10.1016/s0959-4752(99)00029-8

De Corte, E. (2013). Zgodovinski razvoj razumevanja učenja [Theories of learning: A historical approach]. In H. Dumont, D. Istance, & F. Benavides (Eds.), *O naravi učenja: Uporaba raziskav za navdih prakse* [The nature of learning: Using research to inspire practice]. Ljubljana: Zavod RS za šolstvo.

Elstad, E., & Turmo, A. (2010). Students' self-regulation and teacher's influence in science: Interplay between ethnicity and gender. *Research in Science & Technological Education*, *28*, 249–260. https://doi.org/10.1080/02635143.2010.501751

Halinen, I. (2015). *General aspects of basic education curriculum reform 2016 Finland*. Helsinki: Finnish National Board of Education.

Ireland, R. D., Hitt, M., & Sirmon, D. G. (2003). A model of strategic entrepreneurship: The construct and its dimensions. *Journal of Management*, *29*, 963–989. https://doi.org/10.1016/S0149-2063(03)00086-2

Istance, D., & Dumont, H. (2013). Smernice za učna okolja v 21. stoletju [Guidelines for Educational Environment in 21th Century]. In H. Dumont, D. Istance, & F. Benavides (Eds.), *O naravi učenja: Uporaba raziskav za navdih prakse* [The nature of learning: Using research to inspire practice]. Ljubljana: Zavod RS za šolstvo.

Johannisson, B. (2010). In the beginning was entrepreneurship. In B. Frederic, B. Bjerke, & A. W. Johansson (Eds.), *(De) mobilizing the entrepreneurship discourse: Exploring entrepreneurial thinking and action*. Cheltenham: Edward Elgar Publishing.

Jury, J. J. (1999). Focus: Sweden—Venture capitalists flock to Swedish high-tech. *European Venture Capital Journal*, *1*, 34–38.

Kilpatrick, S., Field, J., & Falk, I. (2003). Social capital: An analytical tool for exploring lifelong learning and community development. *British Education Research Journal*, *29*, 417–433. https://doi.org/10.1080/01411920301859

Klingebiel, R., & Adner, R. (2014). Real options logic revisited: The performance effects of alternative resource allocation regimes. *Academy of Management Journal*, *58*, 221–241. https://doi.org/10.5465/amj.2012.0703

Kolovelonis, A., Goudas, M., & Dermitzaki, I. (2011). The effect of different goals and self-recording on self-regulation of learning a motor skill in a physical education setting. *Learning and Instruction*, *21*, 355–364. https://doi.org/10.1016/j.learninstruc.2010.04.001

Kucel, A., Róbert, P., Buil, M., & Masferrer, N. (2016). Entrepreneurial skills and education-job matching of higher education graduates. *European Journal of Education*, *51*, 73–89. https://doi.org/10.1111/ejed.12161

Löbler, H. (2006). Learning entrepreneurship from a constructivist perspective. *Technology Analysis & Strategic Management*, *18*, 19–38. https://doi.org/10.1080/09537320500520460

McGrath, R., & MacMillan, I. (2000). *The entrepreneurial mindset*. Boston, MA: Harvard Business School Press.

Meinel, C., & Leifer, L. (2011). Design thinking research. In H. Plattner, C. Meinel, & L. Leifer (Eds.), *Design thinking: Understand - improve - apply*. Heidelberg: Springer Verlag.

Murgatroyd, S. (2010). 'Wicked problems' and the work of the school. *European Journal of Education*, *45*, 259–279. https://doi.org/10.1111/j.1465-3435.2010.01428.x

Neck, H. M., & Greene, P. G. (2011). Entrepreneurship education. Known worlds and new frontiers. *Journal of Small Business Management*, *49*, 55–70. https://doi.org/10.1109/emr.2012.6210514

Niemi, H., Toom, A., & Kallioniemi, A. (Eds.) (2016). *Miracle of education: The principles and practices of teaching and learning in Finnish schools*. Heilderberg: Springer Verlag.

Pollard, V., & Wilson, E. (2013). The "entrepreneurial mindset" in creative and performing arts higher education in Australia. *Artivate: A Journal of Entrepreneurship in the Arts*, *3*, 3–22. https://doi.org/10.1234/artivate.v3i1.67

Rauth, I., Köppen, E., Jobst, B., & Meinel, C. (2010). Design thinking: An educational model towards creative confidence. In T. Taura & Y. Nagai (Eds.), *Proceeding of the 1st International Conference on Design Creativity*. London: Springer, ICDC.

Sahlberg, P., & Oldroyd, D. (2010). Pedagogy for economic competitiveness and sustainable development. *European Journal of Education*, *45*, 280–299. https://doi.org/10.1111/j.1465-3435.2010.01429.x

Sarasvathy, S. D., & Venkataraman, S. (2011). Entrepreneurship as method: Open questions for an entrepreneurial future. *Entrepreneurship Theory and Practice*, *35*, 113–135. https://doi.org/10.1111/j.1540-6520.2010.00425.x

Seikkula-Leino, J., Ruskovaara, E., Ikavalko, M., Mattila, J., & Rytkola, T. (2010). Promoting entrepreneurship education: The role of the teacher. *Education + Training*, *52*, 117–127. https://doi.org/10.1108/00400911011027716

Shane, S., & Venkataraman, S. (2000). The promise of entrepreneurship as a field of research. *Academy of Management Review*, *25*, 217–236. https://doi.org/10.1007/3-540-48543-0_8

Strauss, A., & Corbin, J. M. (1997). *Grounded theory in practice*. Thousand Oaks, CA: SAGE Publications.

van Merriënboer, J. J. G., McKenney, S., Cullinan, D., & Heuer, J. (2017). Aligning pedagogy with physical learning spaces. *European Journal of Education*, *52*, 253–267. https://doi.org/10.1111/ejed.12225

Žakelj, A., & Ivanuš Grmek, M. (2013). Ability grouping and pupils' results on the national assessment of knowledge. *Croatian Journal of Education*, *15*, 439–463.

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